

Beyond Residuals: A Progressive Semantic-Preserving Quantization Approach for Recommendation

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Abstract. Discrete semantic IDs generated by quantization techniques are becoming essential for enabling efficient and effective large-scale recommendation systems. However, prevailing methods based on Residual Quantization often suffer from two critical limitations: 1) *semantic degradation*, where the core meaning of an item is progressively lost in deeper quantization layers, and 2) the *hourglass phenomenon*, where an imbalanced codebook distribution leads to information bottlenecks and restricts the model’s representational capacity. To address these issues, we propose Dense Residual Quantization, a novel framework that preserves semantic integrity throughout the quantization process. The core innovation of DenseRQ lies in reincorporating the original content embedding at each quantization layer by concatenating it with the corresponding residual vector. This approach anchors fine-grained detail learning to the item’s core semantics, thereby preventing semantic degradation and enabling nuanced item differentiation. Extensive experiments on public and large-scale industrial datasets demonstrate that DenseRQ significantly outperforms state-of-the-art methods in downstream recommendation tasks. Furthermore, online A/B tests validates the practical effectiveness of DenseRQ, yielding substantial improvements in key business metrics. In-depth analysis also reveals that DenseRQ generates a more balanced and expressive codebook distribution, effectively mitigating the aforementioned bottlenecks.

Keywords: Music Recommendation · Quantization · Semantic ID

1 INTRODUCTION

A promising direction for item representation learning is the generation of discrete semantic IDs via Vector Quantization (VQ)[5, 16, 7]. This approach maps high-dimensional item embeddings into a sequence of discrete codes[26, 6, 18],

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providing significant benefits in both efficiency and compatibility with generative models. In particular, Residual Quantization (RQ)[12, 30, 14] has been widely adopted in recent recommendation systems. Methods such as RQ-Kmeans[1] and RQ-VAE[17, 15, 23] utilize a multi-layer architecture, where each layer quantizes the residual error from the preceding layer, thereby enabling more detailed and fine-grained item representations.

However, existing RQ-based methods face two critical challenges. (1) *semantic degradation*: deeper quantization layers cause residuals to drift from core semantics of the item. For RQ-Kmeans, first-layer clusters align with consistent themes (e.g., *pop music*), but second-layer residual clusters group semantically disparate items (e.g., unrelated genres), harming recommendation performance. PSRQ [25] avoids this drift but suffers from overly similar first- and second-layer clusters, limiting semantic ID diversity. (2) *hourglass phenomenon*, prominent in RQ-VAE: imbalanced codebook usage creates an information bottleneck. While the semantic IDs of first-layer often map one-to-one with items, a few semantic IDs of second-layer concentrate hundreds of thousands of items out of millions. The Gini coefficient quantifies the severe hourglass phenomenon in RQ-VAE, which scores an extremely high 0.7529 on the industrial dataset, far exceeding RQ-Kmeans (0.2112), PSRQ (0.2951) and DenseRQ(0.2541), as illustrated in Table 4.

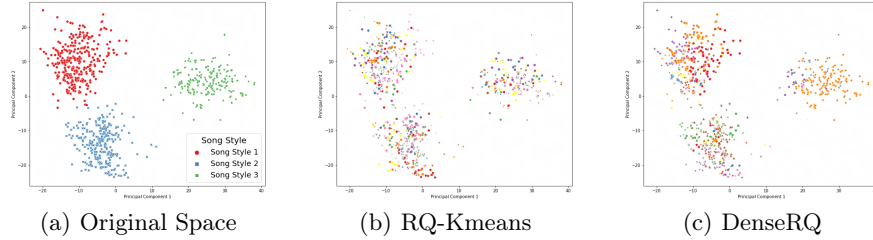


Fig. 1. The subfigure (a) illustrates the original representation space of songs, visualized by PCA. Subfigure (b) displays the clustering results of the second layer in RQ-Kmeans. Subfigure (c) presents the clustering results of the second layer using the proposed DenseRQ method.

To address the challenges of semantic degradation and imbalanced codebook distribution, we propose Dense Residual Quantization(DenseRQ). We argue that **original content embedding contains the most semantic information of item**, which plays vital roles in residual quantization. Thus the design principle of DenseRQ is to preserve original semantic information across all quantization layers. Specifically, at each layer, DenseRQ concatenates the residual vector with the original item embedding before feeding it to the next layer for K-means[19] clustering step. This re-injection of original semantics directly mitigates the degradation observed in RQ-Kmeans. By combining the residual (new information) with the original content embedding (stable context), it also avoids

the overly-consistent clusters of PSRQ. Consequently, this balanced approach allows for the capture of fine-grained details on a stable semantic foundation, effectively mitigating the *hourglass phenomenon* seen in VAE-based methods.

Our contributions can be summarized as follows:

- We conduct a systematic analysis of two critical limitations in existing residual quantization methods for recommendation: semantic degradation and the *hourglass phenomenon*, revealing their adverse effects on the quality and expressiveness of semantic IDs.
- We propose DenseRQ, a novel quantization framework that mitigates semantic degradation by densely re-injecting the original item embedding at each quantization layer. This design preserves semantic coherence across layers while enabling fine-grained residual learning, effectively balancing semantic preservation and diversity.
- Through extensive offline experiments on public and industrial datasets, as well as online A/B tests, we demonstrate that DenseRQ consistently outperforms state-of-the-art methods across multiple recommendation tasks, validating its effectiveness and practical utility in industrial scenarios.

2 RELATED WORK

2.1 Semantic ID Generation Methods

Generating multi-layer semantic IDs from high-dimensional embeddings is a key research challenge. Using Residual Quantization VAE (RQ-VAE) was found to suffer from an *hourglass phenomenon* [11], leading to imbalanced codebook usage. Subsequent solutions drew from NLP, using techniques like N-gram and SentencePiece Modeling (SPM) to create more balanced codebooks [23, 28]. A parallel research direction focuses on the core quantizer, where studies show RQ-Kmeans generally outperforms RQ-VAE [9]. However, RQ-Kmeans suffers from semantic degradation, as deeper-layer residuals lose original semantic context. An attempt to address this issue was made with Progressive Semantic Residual Quantization (PSRQ) [25].

2.2 Application of Semantic IDs in Recommender Systems

The success of quantization technology in domains such as computer vision[13, 24] has inspired its application within recommender systems, enabling novel modeling paradigms for both recall and ranking. VQ-Rec[5] was a foundational work in this area, utilizing product quantization (PQ)[8] in item text representations to learn discrete item codes for recommendation tasks. A significant change occurred with the advent of generative recommendation models that treat recommendation as a sequence generation problem. TIGER[20] was a pioneering framework that leveraged semantic IDs for this purpose. The performance of such downstream models is intrinsically linked to the quality of the semantic ID.

3 METHODOLOGY

3.1 Residual Quantization

Residual Quantization With K-means RQ-Kmeans is a typical residual quantization algorithm that sequentially reduces quantization error. The first layer applies K-means to the original content embedding to generate an initial codebook. Each subsequent layer then applies K-means to the residual vectors from the preceding layer, creating a new codebook to model the remaining error. Finally, a vector is encoded as a tuple of indices pointing to codewords in each codebook. Figure 2 illustrates the working process of RQ-Kmeans.

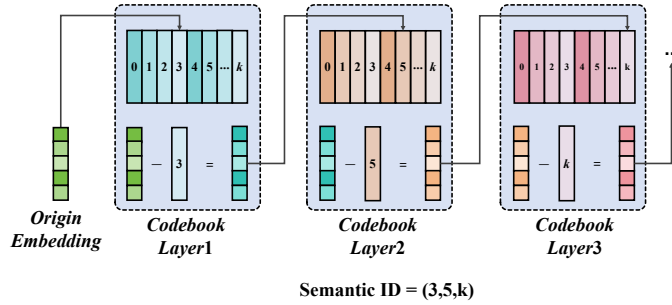


Fig. 2. Illustration of the semantic ID generation process with the RQ-Kmeans method.

Residual Quantized Variational Autoencoder RQ-VAE is a multi-layer vector quantization algorithm integrated within a Variational Autoencoder. In contrast to RQ-Kmeans, the codebooks are not learned via a separate clustering algorithm but are optimized in an end-to-end fashion as learnable parameters. The total loss function $\mathcal{L}_{\text{total}}$ consists of a reconstruction term $\mathcal{L}_{\text{recon}}$ and a quantization term $\mathcal{L}_{\text{quant}}$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{quant}} \quad (1)$$

$$\mathcal{L}_{\text{recon}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 \quad (2)$$

$$\mathcal{L}_{\text{quant}} = \|\text{sg}(\mathbf{z}_e) - \mathbf{z}_q\|_2^2 + \beta \|\mathbf{z}_e - \text{sg}(\mathbf{z}_q)\|_2^2 \quad (3)$$

where Eq. 2 is the Mean Squared Error between the input $\mathbf{x}$ and its reconstruction $\hat{\mathbf{x}}$. The quantization loss in Eq. 3 consists of two terms: a codebook loss that optimizes the codebook embeddings, and a commitment loss, weighted by β , that encourages the encoder output $\mathbf{z}_e$ to remain close to the chosen codebook vector $\mathbf{z}_q$. The $\text{sg}(\cdot)$ operator is the stop-gradient.

Progressive Semantic Residual Quantization Both PSRQ and RQ-Kmeans using K-means are multi-layer vector quantization techniques designed to create compact and accurate vector representations. They share the foundational goal of progressively refining an approximation by quantizing the residual vector from

the previous stage. However, they diverge fundamentally in their operational mechanics, particularly in how they calculate and propagate these residual vectors between layers. PSRQ is that the residual vector obtained from each layer is first subtracted from the original content embedding to generate a new residual vector. This new residual vector is then concatenated with the initial residual vector from the very first step, and the resulting combined vector serves as the input for the next layer.

3.2 Dense Residual Quantization

We propose DenseRQ, a novel progressive quantization method that improves upon RQ-Kmeans and PSRQ. Its core mechanism is to concatenate the residual with the original content embedding at each layer. Crucially, the residual vector is partitioned into blocks, which undergo a varying number of refinement steps. This processing ensures that different partitions contribute multi-layer residual information during K-means clustering. As a result, DenseRQ preserves original semantic information while capturing a rich hierarchy of residual information. This process can be expressed by the following equations:

$$\begin{aligned}
X_1 &= X \\
C_1 &= \text{Kmeans}(X_1, k) \\
X_2 &= (X - \text{Nearest}(X_1, C_1)) \oplus X \\
C_2 &= \text{Kmeans}(X_2, k) \\
&\vdots \\
X_l &= (X_{l-1} - \text{Nearest}(X_{l-1}, C_{l-1})) \oplus X \\
C_l &= \text{Kmeans}(X_l, k)
\end{aligned} \tag{4}$$

where l is the number of quantization layers, $C_l \in \mathbb{R}^{k \times (d \times l)}$ are the generated cluster center embeddings of layer l . $X \in \mathbb{R}^{|Z| \times d}$ are original content embeddings, $|Z|$ denotes the number of items, d denotes the dimensionality of original content embedding. $X_l \in \mathbb{R}^{|Z| \times (d \times l)}$, k is the number of cluster centers of K-means, $\oplus$ denotes the concatenation operation, and $\text{Nearest}(\cdot)$ denotes the nearest representation search operation in the cluster center embeddings.

The structure of DenseRQ, as illustrated in Figure 3, is distinct from that of methods like RQ-Kmeans. In DenseRQ, the input vector dimension for each layer l increases linearly with the number of layers. Specifically, at layer l , the input is a concatenation of the residual vector from layer $l - 1$ and the original content embedding. Consequently, by the time the process reaches layer l , the portion of the vector corresponding to the first layer has undergone $l - 1$ residual computations, while the portion from layer i has undergone $l - i$ computations. Therefore, this method, effectively preserves semantic information from both the original content embedding and the residual vectors of each layer.

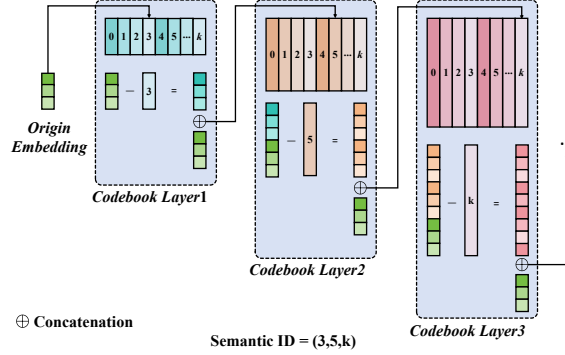


Fig. 3. Illustration of the semantic ID generation process of the DenseRQ method.

3.3 DenseRQ as a Plug-and-Play Module

As shown in Figure 4, DenseRQ as a semantic ID generation method, is independent of the specific type of downstream task. It can be easily integrated into both retrieval models and ranking models of recommender systems.

Combined with the Retrieval Models. In the retrieval stage, the semantic IDs generated by DenseRQ can serve as a direct replacement for representations from quantization methods such as RQ-Kmeans. Following the TIGER framework, a retrieval model can then perform recall by autoregressively predicting the subsequent semantic ID based on a sequence of semantic IDs generated from DenseRQ.

Combined with the Ranking Models. For ranking models, DenseRQ can be employed to enhance user sequence representations. Specifically, each item in the user historical behavior sequence is first mapped to semantic IDs. The embeddings of these semantic IDs are then aggregated by pooling functions such as sum pooling. The output is subsequently fused with the other features of the item, using either concatenation or addition. This approach leverages DenseRQ to provide auxiliary features, thereby enriching the user’s behavioral representation.

4 EXPERIMENTS

In this section, we conduct comprehensive offline and online experiments to evaluate our proposed method. The primary goal of our experimental evaluation is to investigate the following research questions:

- RQ1: How effective is DenseRQ for recommendation tasks in comparison with existing approaches for semantic ID generation?
- RQ2: Does DenseRQ effectively mitigate semantic degradation and the *hour-glass phenomenon* compared to other quantization methods?
- RQ3: How does the proposed method perform in industrial system?

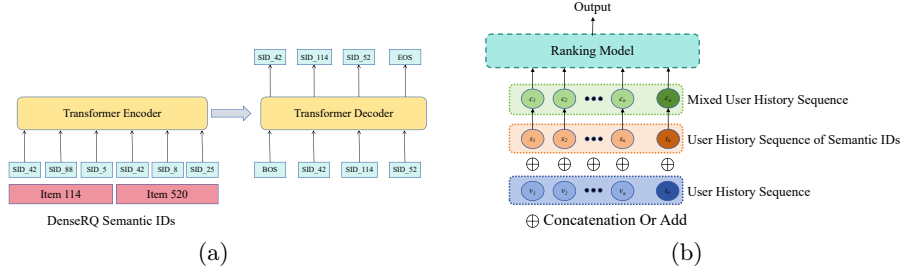


Fig. 4. DenseRQ as a Plug-and-Play Module for Retrieval and Ranking. (a) Application of DenseRQ in the Retrieval Stage. (b) Application of DenseRQ in the Ranking Stage.

4.1 Experimental Setting

Dataset. To evaluate the effectiveness of our method for recommendation systems, we conducted experiments on two public benchmark datasets, Amazon-Baby[4] and Music4All[21], and a proprietary industrial dataset. The industrial dataset was constructed from real-world user logs, capturing strong positive interactions such as full plays, likes, shares and comments. It is formulated for the task of predicting a user’s preference for candidate items based on their historical interaction records. Table 1 presents the number of users, items, and interactions for the three processed datasets. For two public datasets, items with fewer than 30 interactions are marked as cold-start items, while for the industrial dataset, cold-start items refer to newly released songs.

Table 1. Dataset Statistics.

	Amazon-Baby	Music4all	Industrial
#User	0.08M	0.01M	>40M
#Item	0.03M	0.1M	>6M
#Interaction	0.23M	2.6M	>1B

Evaluation Metrics. To evaluate the performance of different methods, we adopt the **AUC** (Area Under the Curve)[3] as the primary evaluation metric. We define **All AUC** as the **AUC** metric evaluated on all items, while **Cold AUC** refers to the **AUC** metric for cold-start items.

4.2 Offline Evaluation

Comparison of Quantization Methods(RQ1). We conduct experiments on two public datasets, Amazon-baby and Music4all, and one large-scale industrial dataset. We use DIN[29] as the ranking model for the public datasets and Climber[27] for the industrial one. Across all settings, we use the Adam[10] optimizer. For a fair comparison, all evaluated quantization methods are configured to generate 3 layers of semantic IDs from 64-dimensional embeddings, with

each layer containing 256 centroids. We compare DenseRQ with the following quantization methods:

- RQ-Kmeans[2]: A standard implementation where K-means is applied sequentially to residuals. We use a fixed random seed to ensure reproducibility.
- PSRQ: PSRQ shares an identical parameter setup with RQ-Kmeans but diverges in its quantization process.
- RQ-VAE: A VAE-based method. Its encoder and decoder are both two-layer MLPs ([64, 64]). To stabilize training, its codebooks are initialized with pre-computed K-means centers.

As shown in Tables 2 and 3, all the tested semantic ID generation methods provided a lift over the baseline, DenseRQ stood out by delivering the largest improvements across every dataset. This superior performance is a direct result of its innovative progressive quantization architecture, which better preserves critical information.

Table 2. Results of the comparison between different quantization methods on two public datasets. Best results are in bold, and second-best are underlined. Improve shows the relative improvement of DenseRQ over the best baseline.

	Amazon-baby			Music4all		
	AUC	Cold AUC	LogLoss	AUC	Cold AUC	LogLoss
DIN	0.6510	0.6009	2.5814	0.7741	0.7887	2.7397
+RQ-Kmeans	<u>0.6553</u>	0.6147	2.5555	0.7765	<u>0.7899</u>	<u>2.7319</u>
+RQ-VAE	0.6548	0.6198	<u>2.5534</u>	0.7719	0.7847	2.7660
+PSRQ	0.6550	<u>0.6207</u>	2.5549	<u>0.7763</u>	0.7874	2.7323
+DenseRQ	0.6571	0.6208	2.5487	0.7770	0.7902	2.7313
%Improve	+0.27%	+0.016%	-0.18%	+0.09%	+0.038%	-0.022%

Table 3. Results of the comparison between different quantization methods on industrial dataset. Best results are in bold, and second-best are underlined. Improve shows the relative improvement of DenseRQ over the best baseline.

Method	AUC	Cold AUC	LogLoss
Climber	<u>0.8363</u>	0.8390	0.9162
+RQ-Kmeans	0.8361	<u>0.8395</u>	<u>0.9158</u>
+RQ-VAE	0.8361	0.8365	0.9261
+PSRQ	0.8357	0.8393	0.9159
+DenseRQ	0.8365	0.8400	0.9153
%Improve	+0.024%	+0.06%	-0.055%

Comparison of Codebook Distributions(RQ2). We evaluate four semantic ID generation methods on a large-scale industrial dataset. To assess the

Table 4. Gini Coefficient of Second-Layer Semantic IDs for Different Methods on Three Datasets, higher GINI Coefficient means more severe *hourglass phenomenon*

	RQ-Kmeans	RQ-VAE	PSRQ	DenseRQ
Industrial	0.2112	0.7529	0.2951	0.2514
Amazon-Baby	0.1998	0.5794	0.3327	0.2445
Music4all	0.1258	0.6952	0.2914	0.2161

disparities in their codebook distributions, we examine two key aspects: (1) the Gini coefficient of the second-layer semantic IDs, which measures the uniformity of the ID distribution, allowing us to evaluate the severity of the *hourglass phenomenon*; and (2) the degree of dissimilarity between the first- and second-layer clustering outcomes, which indicates how residual information is captured, thereby assessing the degree of semantic degradation. All methods are uniformly configured with a three-layers architecture, featuring 256 clusters per layer.

Table 4 shows that DenseRQ achieves a more uniform ID distribution than RQ-VAE and PSRQ. Its superiority over PSRQ stems from its balanced concatenation of the original content embedding and residual vector. The slightly higher Gini coefficient compared to RQ-Kmeans is a deliberate trade-off: to mitigate semantic degradation, DenseRQ re-injects the original content embedding at each layer, which naturally introduces a slight bias toward common IDs. In summary, DenseRQ effectively mitigates the severe *hourglass phenomenon* while making a principled trade-off to preserve semantic integrity.

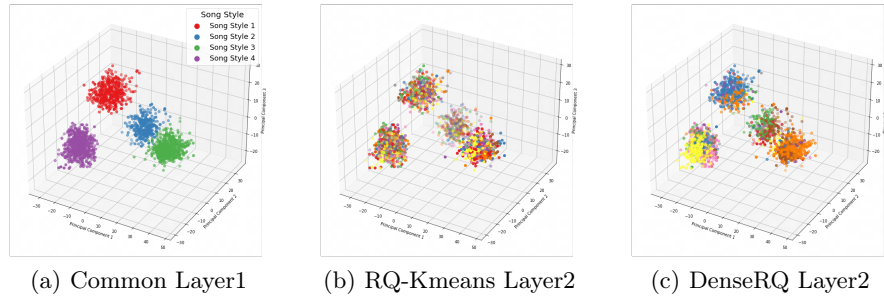


Fig. 5. Visualization of the first and second-layer clustering of some items for semantic IDs, generated by reducing the semantic feature space to three dimensions using PCA.

We visualize the semantic ID clustering structures of different quantization methods by projecting their feature space into three dimensions using PCA[22]. The visualization in Figure 5 reveals critical structural differences:

- RQ-Kmeans: In stark contrast, the second-layer clusters of RQ-Kmeans exhibit almost no correlation with the first layer. This lack of structural continuity is also suboptimal, as it prevents the residual from leveraging the initial groupings to learn relationships among semantically similar items.

- DenseRQ: DenseRQ strikes an effective balance. Its second-layer clustering preserves the foundational structure of the first layer to some extent, while also introducing meaningful divergence. This balanced approach is crucial, as it enables the residual embeddings to learn finer-grained semantic differences built upon the initial semantic similarities.
- RQ-VAE: Unlike the other methods, RQ-VAE does not employ K-means for direct clustering. This fundamental difference leads to its first-layer clustering results being inconsistent with those of the other approaches. Therefore, in this experiment which specifically compares clustering space disparities, the results from RQ-VAE are not directly comparable and thus were excluded from the analysis.

The superiority of this balanced strategy is ultimately validated by the performance metrics on the downstream tasks.

Online A/B Test(RQ3). Online A/B tests were conducted to evaluate the proposed method in two recommendation scenarios of NetEase Cloud Music, with the Climber model serving as the baseline. For the experimental group, DenseRQ was integrated into the Climber model. The primary evaluation metric was the like rate, defined as the proportion of users who collect a track during playback, encompassing both the Overall Like Rate and New Tracks Like Rate. The experimental results are summarized in Table 5. Integrating DenseRQ with the Climber model delivered notable improvements: for the Overall Like Rate, relative increases of 1.597% in Scenario 1 and 0.667% in scenario 2 were observed. The improvement was more striking for the new tracks, where DenseRQ attained performance gains of 3.87% (scenario 1) and 14.08% (scenario 2) compared to the baseline.

Table 5. The relative improvement of Like Rate under different industrial online scenarios.

Scenarios	New Tracks Like Rate	Overall Like Rate
scenario 1	+3.87%	+1.597%
scenario 2	+14.08%	+0.667%

5 CONCLUSION

This work addresses two critical issues in residual quantization for recommendation systems—semantic degradation and the hourglass phenomenon—by proposing DenseRQ, a novel progressive quantization framework. Its core innovation lies in re-injecting original item embeddings at each layer, preserving semantic coherence while capturing fine-grained residual details. Offline experiments on multiple datasets show DenseRQ outperforms RQ-Kmeans, PSRQ, and RQ-VAE in AUC and LogLoss. Online A/B tests validate its practical value, with like rate

of new songs rising up to +14.08% and overall like rate improved. Its balanced codebook, which shows lower Gini coefficient than RQ-VAE/PSRQ, effectively mitigates the hourglass phenomenon. Future research will focus on advanced evaluation metrics for semantic IDs and codebook interpretability.

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